Mohamed Aziz Kammoun

Web Developer · AI System Developer

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Professional Summary

Web developer and AI system developer with hands-on experience building intelligent web applications, automation systems, and gamified learning platforms. Skilled across the complete development lifecycle — from scalable backend architecture with NestJS, Express.js, Django and AI-powered pipelines to animated, immersive frontend experiences with Next.js and React. Passionate about combining AI and real-world impact.

Technical Skills

- **Languages:** TypeScript · JavaScript · Python · Java · SQL
- **Frontend:** Next.js · React · HTML5 · CSS3 · Tailwind CSS · UI/UX Design
- **Backend:** NestJS · Node.js · Django · REST APIs · Java (Native)
- **AI / ML:** NLP · Classification Models · Model Training & Evaluation · AI Automation · AI System Design
- **Data & Scraping:** Web Scraping · Data Preprocessing · Feature Engineering · Markdown Parsing
- **DevOps & Tools:** Git · GitHub Actions (CI/CD) · Docker · Linux (Ubuntu)
- **Other:** Gamification Systems · System Design · Scalable Architecture
- **Hardware & Embedded:** ESP32 · Arduino · Raspberry Pi · RFID Systems · C/C++

Projects

MindFlow — Multi-Agent AI Study Partner

React · Express.js · FastAPI · MongoDB · RabbitMQ · RAG

- Led a team of seven developers to build a study-assistance ecosystem, combining a React interface, eight Node.js microservices, and Python AI agents.
- Integrated course ingestion and retrieval-augmented study planning with scheduling, coaching, evaluation, and camera-based focus and fatigue tracking.

FIN-SHIELD — Fiscal Review & Investigation Platform

Next.js · TypeScript · FastAPI · PostgreSQL · LLM tool use · RAG

- Built a synthetic-data fiscal review platform with explainable risk signals, case workflows, analyst feedback, and role-based access.
- Developed a separate investigation agent with API tools, what-if analysis, document retrieval, and citation verification to produce evidence-grounded drafts for human review.

IEEE Tunisia Partnership Hub — Membership & Sponsorship Platform

Next.js · NestJS · TypeScript · Prisma · PostgreSQL · Better Auth

- Built a full-stack platform for IEEE Tunisia Unit reviews, memberships, invitations, officer positions, partner directories, sponsorship agreements, and support tickets.
- Implemented transactional authorization, audit records, a PostgreSQL email outbox, and a typed API client backed by shared Zod contracts.

Hackership-AI — Internship AI Orchestration

Python · Flask · LiteLLM · Pydantic · RabbitMQ · Redis

- Led a team of developers to build AI workflows for feature decomposition, dependency-ordered task generation, mission grouping, and candidate resume extraction.
- Combined multi-provider LLM routing with structured-output validation, deterministic dependency checks, and asynchronous workers with retries and dead-letter handling.

HealFlow — Data Contract Validation SDK

Python · Pydantic · Pandas · pytest · mypy

- Developed a typed Python SDK that validates Pandas data against versioned contracts before it enters downstream pipeline stages.
- Implemented required-column, type, nullability, range, pattern, and uniqueness checks with structured findings, severity levels, and execution metadata.

Digestible — AI Content Simplification Tool

NLP · Automation · Productivity

- Developed an NLP-powered system transforming complex technical content into readable structured output
- Implemented text summarization, markdown parsing, and structured output generation pipelines

MedAI Nexus — Medical Analysis Prototype

Next.js · NestJS · FastAPI · PostgreSQL · Prisma · scikit-learn

- Led a team of developers to build a medical analysis prototype spanning a Next.js interface, NestJS API, and Python AI service.
- Integrated diabetes, autism survey, and kidney disease prediction models with structured intake forms, AI gateway routing, and confidence-scored results.

SmartBusRFID — Smart Transportation System

Embedded · Real-world Systems

- Developed an RFID-based bus tracking system for real-time user identification and fleet monitoring, and payments via RFID card.

e-Krini — Full-Stack Car Rental Platform

Full-Stack · System Architecture

- Built a complete web platform with frontend/backend integration, database management, and car rental user flows with a simple interface

Internships

Web Developer Intern — Hikma Learn

ESPRIT TECH, Tunisia · July 2025 – August 2025

- Developed **Hikma Learn**, a gamified AI-powered learning platform, as the core internship project.
- Built a React frontend with scroll-driven animations, rank systems, adaptive rewards, progress tracking, and AI-assisted learning features.

Software Development Intern — MindFlow

ESPRIT TECH, Tunisia · July 2026 – August 2026

- Led a team of seven developers to build MindFlow, a multi-agent AI study-assistance platform.
- Coordinated development across the React frontend, Express.js microservices, and Python AI service for study planning, coaching, and progress evaluation.

Software Development Intern — Hackership

Felcloud · July 2026 – August 2026

- Led a developer team building Hackership's internship-management AI with Flask, LiteLLM, and Pydantic, covering task decomposition, mission generation, and resume extraction.
- Implemented taxonomy-based skill extraction and weighted matching scores, with Redis caching for repeated matching requests.

Leadership & Volunteering

Webmaster — RAS Chapter & CS Chapter

IEEE ESPRIT Student Branch, Tunisia · 2 Mandates

- Served two consecutive mandates as Webmaster for the IEEE Robotics & Automation Society (RAS) Chapter and the Computer Society (CS) Chapter
- Managed digital communication, visual content production, and social media presence for both chapters
- Coordinated media campaigns for technical events, workshops, and community outreach initiatives

Volunteer

IEEE ESPRIT Student Branch, Tunisia · Ongoing

- Active volunteer contributing to event organization and community engagement across student branch activities

Education

Engineering Degree in Computer Science

ESPRIT — Ecole Superieure Privee d'Ingenierie et de Technologies, Tunisia · 2024 – Present

Pre-Engineering Cycle

IPEIT — Institut Preparatoire aux Etudes d'Ingenieurs de Tunis, Tunisia · 2022 – 2024

Baccalaureat — Sciences de l'Informatique

Tunisia · 2021

Key Strengths

- AI system design & NLP pipeline development
- Creative and immersive UI/UX design
- System design & architecture
- Fast execution on complex, multi-layer projects
- Full-stack web development (Next.js, NestJS)